

Jonas T. Donnenfield

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RESEARCH INTERESTS

My research reconstructs past ocean and hydroclimate dynamics from marine sediment archives. I develop and apply novel analytical approaches — including laser ablation mass spectrometry, synchrotron imaging, and statistical proxy signal extraction — to recover reliable paleoclimate information from complex depositional environments. This work improves understanding of ocean-atmosphere variability across glacial-interglacial timescales.

EDUCATION

- University of Washington**, Seattle, WA June 2025 – Present
PhD Student in Oceanography
Advisor: Dr. Maureen H. Walczak
- Oregon State University**, Corvallis, OR September 2021 – June 2025
Master of Science in Ocean, Earth, and Atmospheric Sciences
Thesis: *Decoding discharge and diagenesis: A multiproxy investigation of Holocene environmental change in Columbia River-adjacent continental margin sediments*
Advisors: Dr. Maureen H. Walczak and Dr. Alan C. Mix
- Carleton College**, Northfield, MN September 2014 – June 2018
Bachelor of Arts in Geology with distinction, magna cum laude (GPA 3.9)
Thesis: *Recent paleoceanographic history of the San Diego margin, California: A benthic foraminiferal perspective*
Advisors: Dr. Daniel P. Maxbauer (Carleton College) and Dr. Tessa M. Hill (UC Davis)

FELLOWSHIPS, AWARDS, & HONORS

Oregon State University Richard D. Mathew Memorial Oceanography Scholarship (\$500)	2024
Cushman Foundation Loeblich and Tappan Student Research Grant (\$3,000)	2024
PRAX Art+Science Student Fellowship (\$1,000)	2024
Joseph A. Cushman Award for Student Travel (\$1,150)	2022
Geological Society of America Graduate Student Research Grant (\$3,000)	2022
American-Scandinavian Foundation Fellowship (\$23,000)	2022
Oregon State University Provost's Distinguished Graduate Fellowship (\$36,000)	2021–2022
Phi Beta Kappa & Sigma Xi	2018
Carleton College GSA Travel Grant	2018
Carleton College Senior Comps (Thesis) Fellowship (\$2,000)	2017–2018
Sam '75 and Meg Woodside Endowed Fund for Career Exploration (\$4,000)	2017

PUBLICATIONS & PRESENTATIONS

Peer-Reviewed Publications

- Hoogakker, B. A. A., **Donnenfield, J.**, et al., 2025, Reviews and syntheses: Review of proxies for low-oxygen paleoceanographic reconstructions: *Biogeosciences*, 22, 863–957.
- Quintana Krupinski, N. B., Filipsson, H. L., Mackensen, A., Friberg, Y. B., Fanget, A., **Donnenfield, J.**, Knudsen, K. L., Groeneveld, J., Seidenkrantz, M., in preparation, Meltwater release from the Fennoscandian Ice Sheet and potential links with Heinrich Stadial 1 during the last deglaciation.
- Palmer, H. M., Hill, T. M., Roopnarine, P. D., Myhre, S. E., Reyes K., and **Donnenfield, J. T.**, 2020, Southern

California margin benthic foraminiferal assemblages across a modern environmental gradient record recent centennial-scale changes in oxygen minimum zone: *Biogeosciences*, Vol. 17, 2923–2937, doi: 10.5194/bg-17-2923-2020.

Conference Presentations (* = oral)

- ***Donnenfield, J.**, Walczak, M., Mix, A., Fehrenbacher, J., Adler, J., Padman, J., Ross, A., 2025, Holocene Pacific Northwest Hydroclimate and Columbia River Discharge from Continental Margin Sediments: AGU25.
- Donnenfield, J.**, Walczak, M., Mix, A., Fehrenbacher, J., Adler, J., Padman, J., Ross, A., 2025, Assessing Diagenetic Alteration of Foraminiferal Geochemistry in Northeast Pacific Continental Margin Sediments: AGU25.
- Walczak, M., Mix, A., Alder, J., Praetorius, S., Stoner, J., Reilly, B., O'Connor, K., Stelling, K., Cargill, S., **Donnenfield, J.**, 2025, Ocean and climate impacts of southern expansion of the Cordilleran Ice Sheet: AGU25.
- Donnenfield, J. T.**, Walczak, M., Mix, A., Fehrenbacher, J., 2025, Discharge and Diagenesis: A Multiproxy Investigation of Environmental Change in Columbia River-Adjacent Marine Sediment: University of Washington Program on Climate Change Summer Institute.
- Donnenfield, J. T.**, Walczak, M., Mix, A., Fehrenbacher, J., Stoner, J., Cargill, S., Stelling, K., Padman, J., 2024, Climatic drivers of Columbia River discharge and Northeast Pacific oceanography during the Holocene: AGU24.
- Walczak, M., Mix, A., Stoner, J., Fehrenbacher, J., Reilly, B., Stelling, K., **Donnenfield, J. T.**, Cargill, S., 2024, The Beginning of the End of the Marine Cordilleran Ice Sheet: AGU24.
- Donnenfield, J. T.**, Walczak, M., Mix, A., Fehrenbacher, J., Filipsson, H., Padman, J., 2024, Deciphering mechanisms of persistent low oxygen in the Northeast Pacific Ocean over the last 20,000 years: Graduate Climate Conference, Pack Forest, WA.
- Plavetić, M., **Donnenfield, J.**, Choquel, C., Ni, S., Pirzamanbein, B., Filipsson, H., 2024, From the Deep Ocean to Deep Data: Leveraging Advanced μ CT and AI Techniques in Foraminifera Morphological Data Processing, Nordic Geologic Winter Meeting, Gothenburg, Sweden.
- Donnenfield, J. T.**, Cargill, S., Walczak, M., Mix, A., Filipsson, H., Fehrenbacher, J., Padman, J., 2023, Disentangling mechanisms of persistent benthic hypoxia in the NE Pacific from the late Pleistocene to late Holocene: AGU23.
- Hatfield, R., Reilly, B., Klotsko, S., Jennings, A., Mix, A., Stoner, J., Walczak, M., **Donnenfield, J. T.**, Fritz, C., Gregory, E., Kelleher, R., Hough, A., Monito, L., Olarte, P., Stelling, K., Siragusa, M., Vonnahme, T., 2023, The Baffin Bay Deglacial Experiment (BADEX): Spatiotemporal Observations of the Demise of the Last Glacial Maximum Greenland Ice Sheet: AGU23.
- Stelling, K., Fritz, C., Monito, L., **Donnenfield, J. T.**, Walczak, M., Reilly, B., Stoner, J., Hatfield, R., Jennings, A., Klotsko, S., Mix, A., Cheseby, M., Fanshier, C., Frieberg, B., Walczak, P., BADEX Science Team, 2023, Taking CT to Sea: Real-time input from 3D Computed Tomography while coring west Greenland deglacial sediments: AGU23.
- Donnenfield, J. T.**, Stelling, K., Hytrek, H., Fehrenbacher, J., Reilly, B., Praetorius, S., Walczak, M. H., Mix, A., Stoner, J., Filipsson, H., 2023, Northeast Pacific evidence for persistent benthic hypoxia and testing hypothetical linkage to freshwater discharge dynamics from the late Pleistocene to late Holocene: GSA Penrose Conference - The Role of Outburst Floods in Earth and Planetary Evolution, Grand Coulee, WA.
- Donnenfield, J. T.**, Walczak, M. H., Mix, A., Filipsson, H., Fehrenbacher, J., Stelling, K., Gregory, E., 2022, Drivers of margin hypoxia in the late Pleistocene-Holocene North Pacific: 14th International Conference on Paleoceanography, Bergen, Norway.

Donnenfield, J. T., Walczak, M. H., Mix, A., Stelling, K., Hytrek, H., Churchill, A., 2022, Reconstructing benthic hypoxia: A paleoceanographic analogue for the near-future Pacific Northwest margin: 53rd International Liège Colloquium on Ocean Dynamics & 3rd GO2NE Conference, Liège, Belgium.

Donnenfield, J. T., Walczak, M. H., Mix, A., Stelling, K., Hofstetter, C., Gregory, E., Hytrek, H., Dwyer, D., Reilly, B., Stoner, J., Fehrenbacher, J. S., and McKay, J., 2021, Observations of persistent benthic hypoxia on the Pacific Northwest margin: An early Holocene analogue for our near-future: Geological Society of America Abstracts with Programs, Vol 53., No. 6, doi: 10.1130/abs/2021AM-368760.

Donnenfield, J. T., Palmer, H. M., and Hill, T. M., 2018, Recent paleoceanographic history of the San Diego margin, California: A benthic foraminiferal perspective: Geological Society of America Abstracts with Programs, 52nd North-Central, Vol. 50, No. 4, doi: 10.1130/abs/2018NC-312591 (Undergraduate Thesis - full doc on request).

Film & Media

A Map of the World in Time (BADEX expedition film), featured expedition participant; available on [Amazon](#).
Who's Saving the Planet? podcast, co-host (selected episode available on [Spotify](#)).

ACADEMIC APPOINTMENTS

Predoctoral Research Associate June 2025 – Present
University of Washington, Seattle, WA

- Reconstructing Northeast Pacific Ocean biogeochemistry and climate since the Last Glacial Maximum
- Setup and operation of a radiocarbon (^{14}C) preparation laboratory

Research Assistant June 2023 – June 2025
Oregon State University, Corvallis, OR

- 12,000 years of Pacific Northwest climate and Columbia River discharge (Sept 2023 – June 2025)
- Spatiotemporal observations of ice sheet demise from Baffin Bay, West Greenland (June 2023 – Aug 2023)

Teaching Assistant (Geological Oceanography) January 2023 – March 2023
Oregon State University, Corvallis, OR

- Instructed and modified weekly labs, co-led students' final research project, edited and graded assignments

American-Scandinavian Foundation Fellow April 2022 – December 2022
Lund University, Lund, Sweden

- Determining late Pleistocene Baltic Sea history using benthic foraminifera (μCT scanning and 3D modeling)

Provost's Distinguished Graduate Fellow September 2021 – September 2022
Oregon State University, Corvallis, OR

- Investigating ocean dynamics and climate change in the Pacific Ocean/Western US over the past 20,000 years

Teaching Assistant (Paleobiology) March 2018 – June 2018
Carleton College, Northfield, MN

- Taught fossil identification and analyses; facilitated field trips to study evolution of ancient biota and environs

Research Intern July 2017 – September 2017; December 2017
UC Davis Bodega Marine Lab (BML), Bodega, CA

- Investigated ocean deoxygenation through microfossil and isotopic proxies from recent sediment cores

Field Researcher

August 2016 – October 2016

Osservatorio Geologico di Coldigioco, Apiro MC, Italy

- Investigated orbital forcing and paleoclimate of the K-Pg boundary using cyclostratigraphic analysis

Research & Fieldwork Intern

June 2016 – July 2016

San Francisco Estuary Institute (SFEI), Richmond, CA

- Contributed to research on LIDs, sensor-based water quality monitoring, and microplastic contamination

Research Intern

December 2015

UCSF Emotional Health and Psychophysiology Laboratory, San Francisco, CA

- Conducted pilot studies examining physiological response as an index of emotion, motivation, and intention

PROFESSIONAL POSITIONS

Data Scientist

August 2020 – August 2021

United by Zero/Who's Saving the Planet Technologies (WSTP), New York, NY

- Developed sustainability scoring system based on UN SDGs using ensemble learning; presented on company podcast

Community Science Instructor

February 2020 – March 2020; April 2021 – June 2021

Seattle Aquarium, Seattle, WA

- Developed marine science curriculum for underrepresented high school students with hands-on field research

Lab Technician

December 2019 – March 2020

United States Geological Survey, Tacoma, WA

- Identified microplastic polymers using FTIR spectroscopy; developed SOPs for future USGS microplastic studies

Academic Tutor

June 2018 – December 2018

Tutoring and Learning Center, Fairfax, CA

- Taught math, science, reading, and writing to middle and high school students in one-on-one and group sessions

MENTORSHIP

Alanna Magers

January 2024 – June 2024

*Undergraduate Research Assistant, Oregon State University
Sedimentology, CT imagery, micropaleontology***Ethan Audia**

January 2026 – Present

*Undergraduate Research Assistant, University of Washington
Micropaleontology***TRAINING & SKILLS**

Programming Experience

- Python (LATools), R (Bchron), MATLAB (SedCT), Java, LaTeX, ArcGIS, Google Earth Pro, FIJI, Dragonfly, OxCal, IgPet, Stereonet 9, AcqKnowledge, Adobe (Illustrator, Photoshop, Premiere Pro)

Scientific & Related Techniques**Analytical Geochemistry & Laboratory**

- **Organic Geochemistry:** Lipid biomarker extraction and purification (e.g., alkenones, GDGTs) using ASE, column chromatography, GC-FID
- **Mass Spectrometry:** ICP-MS, LA-ICP-MS, ICP-OES, and AMS (¹⁴C sample preparation)
- **Imaging:** Synchrotron-based micro-CT scanning, 3D data processing, and petrographic thin section preparation/mineral identification
- **Micropaleontology:** Taxonomic identification of foraminifera from the North Pacific Ocean and Baltic Sea

Field & Marine Operations

- **Shipboard Operations:** Deployment/recovery of coring mechanisms (Jumbo Piston Core, gravity corer, box corer, GEMAX, Multicorer), CTDs, benthic tripods, moorings, water sensors, and plankton tows
- **Sample Processing:** Field-based processing of sediment and water samples for geochemical and micropaleontological analysis; speleothem sampling
- **Geologic Field Skills:** Mapping, feature orientation measurements, and detailed lithologic/stratigraphic descriptions
- **Biological Handling:** Coastal and intertidal organism capture and handling

FIELD EXPERIENCE

Shipboard

S/V Statsraad Lehmkuhl – NE Pacific coast One Ocean Expedition (UN Ocean Decade)	Oct 2025
R/V Sikuliaq – Oregon continental shelf Gravity coring, multicoring, benthic tripods, subsurface moorings (Cascadia Canyons)	March 2024
R/V Armstrong – Baffin Bay, Greenland Jumbo piston coring, CTDs, plankton tows, water sampling (BADEX)	July – Aug 2023
R/V Alice – Gullmar Fjord, Sweden GEMAX core recovery, CTD casts, water sampling (Stress3D)	Sept 2022
R/V Alice – Gullmar Fjord, Sweden GEMAX & box core recovery, plankton tows, CTD casts, water sampling (FOODIE)	May 2022
R/V Elakha – Yaquina Bay & River, OR Estuarine water chemistry – 10th year of ongoing long-term monitoring	Sept 2021
R/V Oceanus – Oregon continental shelf Multicore recovery, plankton tows, CTD casts (PacHEAT)	Aug 2021
R/V Mussel Point – Bodega Bay, CA Plankton tows and water chemistry for coastal environments	Oct 2017
F/V via SFEI – San Francisco Bay, CA Water sensor deployment and recovery (Regional Monitoring Program)	July 2016

Terrestrial

Oregon Caves National Monument, OR Holocene paleoclimate	July 2024
Channeled Scablands, WA Outburst floods and landscape evolution	June 2023
Skania & Vomb basin, Sweden Quaternary glacial geology and human activity	April 2022

Volcanoes National Park, HI Volcanology and hotspots	March 2018
Death Valley & Mojave Desert, CA Sedimentology and structure of dunes/canyons	Oct 2017
Italy & Croatia Mapping, sedimentology, paleontology, riverine geomorphology	Aug – Oct 2017
Northern MN Mid-continent Rift, LIPs, LGM glacial/deglacial deposits	April 2017
Painted Canyon, CA Structural geology and field mapping of San Andreas fault	Feb 2017
Baraboo, WI Igneous petrology and emplacement of regional quartzite	May 2016
Black Hills & Badlands, SD Petrology of orogeny, geomorphology of paleosols	Oct 2015

SHORT COURSES, WORKSHOPS, & SUMMER SCHOOLS

Carbon Removal in Washington State and Beyond (Buerk Center/Microsoft Climate Innovation Fund)	Jan 2026
University of Washington Program on Climate Change Summer Institute	Sept 2025
Urbino Summer School on Paleoclimatology (USSP)	July 2025
Pages PO2 Workshop	Nov 2024
SynchroMage: 3D Tomography and Visualisation	Oct 2024
Marine Heatwaves Training Workshop	Sept 2024
paleoCAMP	June 2023
Synthesis of Cenozoic paleoceanographic proxies for seawater oxygenation (at ICP14)	Sept 2022
Paleo Data Assimilation Workshop	Aug 2022

VOLUNTEERING & OUTREACH

American Geophysical Union (AGU) Paleoclimatology & Paleoclimatology Early Career Working Group (Paleo ² ECRS)	Jan 2023 – Present
American Geophysical Union (AGU) Fall Meeting 2024 Session Chair (Oxygenation Dynamics in Past Oceans)	Dec 2024
Oregon State University CEOAS Diversity, Equity, and Inclusion Committee (DEI)	Oct 2021 – June 2025
Oregon State University CEOAS Association of Graduate Students (CAGS)	Jan 2022 – June 2022
Oregon State University CEOAS Graduate-Undergraduate Inclusive Development Experience (GUIDE)	Oct 2021 – June 2024